CPSC 490- Undergraduate Seminar in Computer Science

INSTRUCTOR:

Marc Velasco

CONTACT INFORMATION:

Email: mavelasco@fullerton.edu
Discord: https://discord.gg/Zdd4wRBvWr

- When you join make sure you drop your name and section number in General or via DM so I can add you to the proper role, each class has their own discord channel.

COURSE SCHEDULE

Class: Tuesday/Thursday 11:30am-12:45pm, GH246

OFFICE HOURS:

Office Hours: Tuesday/Thursday 1pm-3pm on Zoom by Appt (reach out to me when want to join)
Otherwise: Friday afternoons By Appt on Zoom (reach out to me when want to join)

COURSE DESCRIPTION AND PURPOSE:

Per the University Catalog description:
Review of foundational computer science theories and principles, real-world application development methods and processes, and industry practices. Survey of modern computing technologies. Research methods. Identification of research or practical application problems. Writing and presenting a proposal for a capstone project.

STUDENT LEARNING OBJECTIVES

- Knowledge of computer science theories and principles and application to real-world problems
- Knowledge of industry practices, tooling and processes.
- A survey of modern topics in the industry.
- Research methodology for academic and industry applications
- How to identify opportunities for improvement from research or industry work.
- Creating and presenting proposals for work and or research.

CLASS FORMAT

To summarize and set expectations, this class will entail the establishments of groups for student work, each group will select and work on a project throughout the semester. Students will be evaluated individually on their **contribution** and **competency** as the main contributors to their earned grade.

- Groups will consist of between 2 and no more than 4 students, groups will be setup by me. If needed, I'll move students around to ensure proper number of students per group.
- Groups will decide the idea/purpose and technical functionality of the project with my approval or changes needed to ensure proper complexity and scope of work for 490 and 491
- All groups must be made and formed in the same section, **no** cross section/class groups.
- **No** individual projects or supervised work with another section/professor.
- All students are graded individually
- There will be **no** "incompletes" given out as grades, **no** makeup for missed meetings.
- My sections of 491 will be similar in format, all groups will be in class in the same section, **no** cross-section/class/professor groups, if needed you'll reform or join a new group in 491 as needed.

TEXTBOOKS AND RESOURCE MATERIALS:

Required
Note: These tools must be used to be assigned credit and be graded in the course, any deviation of failure to use these tools will result in no or a failing grade.

- **Google Docs/Slides must be used** for all submissions for documents and slides so I can track history and contributions made per group member.
- **Github must be used** for all code so I can view all code contributions, frequency, and type of contributions.

Suggestions and Recommendations

- No textbook is required.
- Suggested reading materials if any will be posted on canvas.
- Recommended textbooks: =
- *Docs for developers: an engineer's field guide to technical writing*
  Bhatti, Jared, author.; Hightower, Kelsey
  2021
- *Writing for engineering and science students: staking your claim*
  Rau, Gerald, 1954- author.
  2020
- Other tools and technology as needed for groups to meet requirements outlined in the syllabus

- A computer with a web browser and Internet access to the course website

COURSE GRADING:

Grades will be assigned as follows:

A+ 97–100%
A 93–96%
A− 90–92%
B+ 87–89%
B 83–86%
B− 80–82%
C+ 77–79%
C 73–76%
C- 70–72%
D+ 67–69%
D 63–66%
D- 60–62%
F 0–59%

Rounding: I do round up to the nearest single percent, only if you have not missed any presentations.

Note: While I make every effort to ensure accurate grading all work, canvas is not the official gradebook, make sure you are monitoring your grade and correct calculation of your weighted grades based on the table below, my final grades will be based on the weight below for all work.

WHAT CONSTITUTES YOUR GRADE:

The course will have three submissions, which will contribute points towards the following weighted areas of competency for the course:

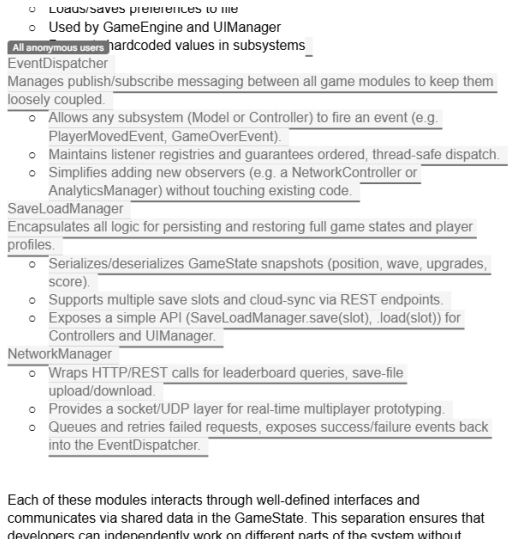
Capstone Pitch and Presentation	10%
Requirements Engineering	20%
Project Planning	20%
Architecture and Design	20%
UX Design	20%
Prototyping	10%

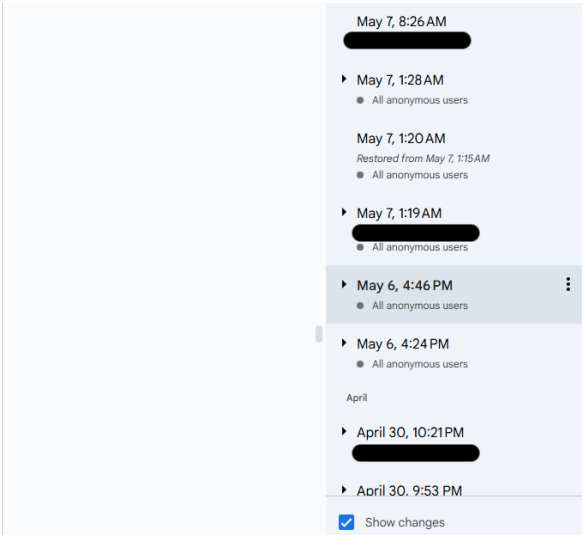
Note: as shown below in sample grading scenarios, the second submission is worth 54% of your total weighted points, if you have not earned 12.5% of weighted points by the end of the first submission there is no mathematical way to pass the class to satisfy the upper division writing requirement.

All students are expected to show evidence of work contributed individually to earn a grade in the assignments in this class. There is no “group grade”.

If no evidence is supplied, no grade or points will be given, this especially includes **anonymous** or unsigned in contributions, work that was done on personal devices but never contributed to GitHub or google docs/sheets etc.

Example:





Each student will need to demonstrate knowledge and competency in various rubric areas in an assignment to get points. Not demonstrating contributions or competency in a rubric area will not earn points to the areas above in "What Constitutes your Grade".

Note: You will earn points for each submission, you cannot make up points once they are gone. Each student is responsible for ensuring they are monitoring their own progress and grades and if they find themselves in a position from which they cannot mathematically pass the class they should make use of all university tools/resources, there will be NO make up assignments/work or extra credit etc.

COMPETENCY AREAS AND POINTS BY SUBMISSION

	Submission 1	**Submission 2**
Project Planning/Metrics		
Project Planning	0-40pts	0pts
Metrics	0-40pts	0-20pts
Requirements Engineering		
Use-Cases	0-40pts	0pts
Modeling	0-40pts	0pts
Architecture and Design	0 pts	0-50pts
UX Design	0 pts	0-40pts

Prototype	0 pts	0-20pts
Total:	160pts	130pts

EXAMPLE GRADING SCENARIOS

STUDENT 1 SUBMITS LAST MINUTE EDITS FOR ARCH/DESIGN (WITHIN LAST WEEK OF CLASS), BUT SUBMITS WORK FOR ALL OTHER SECTIONS:

Capstone pitch: 100/100 = 100%
Project Planning: 100/100 = 100%
Requirements Engineering: 80/80 = 100%
Architecture and Design: 10/50 = 20%
UX Design: 40/40 = 100%
Prototype: 20/20 = 100%

(100* .10 for pitch) +(100*.20 project planning) + (100*.20 req engineering) + (20*.20 arch&design) + (100*.20 UX Design) + (100*.10 prototype) = 84%

STUDENT 2 SUBMITS ALL WORK BUT MISSES 2 PROJECT MEETINGS IN DECEMBER:

Capstone pitch: 100/100 = 100%
Project Planning: 96/100 = 96%
Requirements Engineering: 80/80 = 100%
Architecture and Design: 40/50 = 80%
UX Design: 32/40 = 80%
Prototype: 16/20 = 80%

(100* .10 for pitch) +(96*.20 project planning) + (100*.20 req engineering) + (80*.20 arch&design) + (80*.20 UX Design) + (80*.10 prototype) = 89.2%

STUDENT 3 SUBMITS ALL WORK BUT MISSES THE SUBMISSION 1 PRESENTATION:

Capstone pitch: 100/100 = 100%
Project Planning: 72/100 = 72%
Requirements Engineering: 52/80= 76.67%
Architecture and Design: 50/50 = 100%
UX Design: 26/40 = 65%
Prototype: 13/20 = 65%

(100* .10 for pitch) +(72*.20 project planning) + (52*.20 req engineering) + (100*.20 arch&design) + (100*.20 UX Design) + (100*.10 prototype) = 84.8

STUDENT 4 SUBMITS ALL WORK BUT MISSES THE SUBMISSION 2 PRESENTATION:

Capstone pitch: 100/100 = 100%
Project Planning: 96/100 = 96%
Requirements Engineering: 80/80 = 93.33%
Architecture and Design: 40/50 = 80%
UX Design: 32/40 = 80%
Prototype: 16/20 = 80%

(100* .10 for pitch) +(96*.20 project planning) + (100*.20 req engineering) + (80*.20 arch&design) + (80*.20 UX Design) + (80*.10 prototype) = 89.2%

ATTENDANCE AND PARTICIPATION

GROUP MEETINGS

You will be required to meet with me throughout the semester to keep accountable to the project. Meetings will be in your assigned groups.

Until the Capstone Pitch Presentations are complete, we will meet every week after which we will move to a cadence of roughly every two weeks (I will assign a schedule for groups to meet with half in one week and the other half in a different week).

For project meetings I will take attendance for the group members, students will be afforded one excused absence for the period leading up to each submission (so 1 absence before the capstone pitch, one before submission1, and one before submission 2).

Once your excused absences are used up, any additional absences will trigger a 20% penalty to your total points for any work submitted for that submission. (this is separate from any penalty from missing a presentation)

Note: Students should consider the assigned class time as time assigned for class work, meetings, and presentations, the schedule is tentative and subject to change and if an assigned meeting or presentation date needs to change it will move to another assigned class day and students would still be expected to attend.

PRESENTATIONS

Project presentations will be required three times during the class. A student is responsible for personally presenting as part of a group 5 minutes in which each student must speak live in front of the classroom and sharing via the room AV system.

Every student MUST attend all designated presentation days, even if you have given/presented already, all students are expected and graded on attending every single presentation from all groups.

If you are not in attendance for presentation days:

- For the capstone pitch presentation - student will receive a 15% penalty to your total points earned from the capstone pitch.
- For the first capstone submission presentation - student will receive a 35% penalty to your total points earned for submission 1.
- For the second capstone submission presentation – student will receive a 20% penalty to your total points earned for submission 2.

Students who are late to class on a presentation day will receive a penalty up to the maximum for a missed presentation depending on how late they are to the class session.

- 0-10 minutes late = 5% penalty
- 10-30 minutes late = 15% penalty
- 30+ minutes late = max penalty for missing a presentation (25% for submission 1, 35% for submission 2)

Note: Students should consider the assigned class time as time assigned for class work, meetings, and presentations, the schedule is tentative and subject to change and if an assigned meeting or presentation date needs to change it will move to another assigned class day and students would still be expected to attend.

CAPSTONE PROJECT PITCH AND PRESENTATION

Your group will identify and select a project to work on. Your group will decide what that project is, and who will do what in terms of work. Some guidelines to help decide what to work on:

- Every student will need to show competency in the required areas of submission 1&2 below.
- Every student will need to show hands on keyboard work for the project in 491.

Note every student must show, for grading, **contribution** and **competency**.

The group will deliver for this (each group member must present some part of this):

- A presentation of the idea
- Why it's important
- Why it should be worked on and represents a breadth of work from the Computer Science discipline
- An estimate of the work needed (rough timeline, milestones, who will work on what)
- The rubric and content for this will be as follows:
    - 25% quality of proposal
    - 25% significance and complexity
    - 25% proposed timeline/milestones
    - 25% proposed mockup of project

Note your groups must meet with me to get approval of your project idea for inclusion in the pitch.

CAPSTONE PROJECT SUBMISSIONS AND PRESENTATIONS

You will then proceed to begin developing the capstone proposal and work on the aspects of the idea. This will vary according to the idea, but will need to show architecture, requirements design and modeling, prototyping, proposed development methodologies, POCs etc.

Important: For all the sections below, each student will be individually graded on their **competency** and **contribution** to every section as outlined above.

The project will be broken up into two submissions as follows:

SUBMISSION 1:

- Well-formed abstract (**1 paragraph**)
- Table of Contents (**minimum of 1 page**)
- Introduction (**minimum of 2 pages**)
- Project Plan (**minimum of 3 pages single-spaced with diagrams**)
    - What work will be done, milestones, high-level goals, timeline
- Metrics (**minimum of 3 pages single-spaced with diagrams**)
    - Project tracking metrics, identify metrics to track for the project to show it on schedule/behind schedule, planned vs. actual plan etc., show examples of what graphs/charts you would show for these metrics, identify metrics for 490 and 491
- Requirements engineering (**minimum of 4 pages single-spaced with diagrams per group member**)
    - Use-Cases (not diagrams) in a well defined format like Alistair Cockburn format, showing actors, preconditions, postconditions, steps, error flows etc.
    - Modeling (at least 1 state, 1 use-case diagram, 1 additional behavioral diagram (not state or use-case), and at least 1 story board – comic book/manga depiction of the app being used, how it's used, where it's used etc.)

SUBMISSION 2:

- Add on to the same doc from submission 1: Everything complete from Submission 1, if it wasn't complete it will need to be complete in submission 2.
    - Metrics (**minimum of 3 pages single-spaced with diagrams**)
        - Now you will show real data for metrics throughout 490, by submission2 you should show metrics to show how each member contributed work, who did what etc. from submission 1 through submission 2
    - Architecture and design (**minimum of 3 pages per group member, single-spaced with diagrams**)
        - Cite and identify an architecture pattern to use and a design pattern (note they are different)
        - Provide textual as well as detailed diagrams (original drawings, not recycled diagrams) for your architecture and design choices
        - Justify your selections, why are they appropriate for your project
    - UX Design (**minimum of 2 pages per group member single-spaced with diagrams**, all of the below items must be present)
        - Aesthetic – what aesthetic is being used, selected and exhibited
        - User research – how do you know who your users are, how they will use your application
        - Personas – create user personas to represent different stake holders in the system
        - UX frameworks – what UX frameworks will you use

        - UX architecture and design – what arch and design will you use for UX? Cite and identify patterns used
        - UX Mockup – static picture of the final application
    - References (**minimum of 2 references per group member** in APA format and cited throughout the paper)
    - Prototype (**minimum of 2 pages per group member**) – an animated (responds to clicks/shows flow from one screen to another) demo of part of the application, this should not be the same as the mockup, prototype is a proof of concept of the idea, not the final idea itself.
        - You will submit screenshots of the prototype and a video link (shared link, unlisted youtube video etc.), the prototype can be developed as a code or a mockup (more detailed than the UX mockup that shows just the UX) that shows the flow of the application, difference between this and a mockup, is the prototype will respond to user-interaction (click to next page/screen etc.).

CAPSTONE GRADING AREAS AND EXPECTATIONS

In 490 you will be expected to work on all aspects of the five competencies laid out. Each competency has a maximum number of points available per submission, and not all competencies are available in all submissions. The goal for 490 is to have sustained contribution to the project throughout the class. While not all work has to be done every single day, a healthy sustained contribution throughout the class is the goal.

EXAMPLE RUBRICS FOR THE SUBMISSIONS ARE AS FOLLOWS:

PROJECT PLANNING/METRICS

PROJECT PLANNING

0pts - No project plan evidenced.
5pts - High level milestones or missing work planned for 490 or 491. All work uploaded in one upload/at one time showing very low complexity.
15pts - Terse descriptions of work, or no mention of how resources will be spent (how many are needed for work items). Some work uploaded last minute/ little sustained work throughout class showing middle range complexity.
20pts - Minimally describe milestones identified between 490 and 491 or technical heavy steps before 491. Sustained work throughout class.
30pts - Well documented project plan with clear objectives, timelines, allocation of resources the coveres 490 and 491. Well thought out complexity and sophistication.

METRICS

0pts - No data to show, example data still shown.
5pts - Some examples still used, or inconclusive data shown that ties to work performed.
10pts - Metrics and graphs show some data for part of submission 1 to submission 2. Metrics align with stated work and evidenced work from google doc.
20pts - Metrics show usage and tracking of real data for 490. Metrics and graphs show accurate and up to date information at least from submission 1 to submission 2.

REQUIREMENTS ENGINEERING

USE-CASES

0pts - No use-cases identified or evidenced.
10pts - Requirements and/or non-functional requirements are present. All work uploaded in one upload/at one time showing very low complexity.
25pts - Use-cases are written in a generic non-well-defined format, steps or conditions or flows are ambiguous, or technical details are present as opposed to a non-technical description of what/how the application will be used by end-users. Some work uploaded last minute/ little sustained work throughout class showing middle range complexity.
40pts - Use-cases are written in a well-defined format some ambiguity exists in steps and /or pre/post/error conditions. No technical implementation details are included. Sustained work throughout class.

MODELING

0pts - No modelling at all evidenced.
10pts - Not all minimal number of diagrams are presented. All work uploaded in one upload/at one time showing very low complexity.
20pts - Minimum number of diagrams are presented, 1 state, 1 use-case, 1 behavioral (of any type), with minimal details and concepts (flow, sequence of events, state etc.). Some work uploaded last minute/ little sustained work throughout class showing middle range complexity. Contributions in a small timeframe (work done week before due date as an example).
30pts - Diagrams are present and show good detailed concepts (flow, sequence of events, state etc.). Sustained work throughout class (No less than two weeks of sustained contributions)
40pts - Well detailed and formed diagrams are present, show relation to the use-cases, and display non-trivial requirements concepts (flow, sequence of events, state etc.). Well thought out complexity and sophistication.

ARCHITECTURE AND DESIGN

0pts - No architecture or no design work has been done or evidenced. Minimal or no contributions evidenced, anonymous contributions attributed to student.
10pts - Only an architecture pattern is selected, or only a design pattern is selected, minimal justification or just a statement of the pattern itself, no or little explanation of how it will be used in this project. Contributions in a small timeframe (work done week before due date as an example).

30pts - Architecture and Design patterns selected with minimal description and/or visual description as to why or how it fits and adapts into the project. Infrequent contributions (no less than two weeks of sustained contributions) throughout Submission 1 - Submission 2 timeframe.
40pts - An architecture pattern and design patterns are identified, exhibited via non-generic diagrams (diagrams and terms are specific to the application, not generic stock diagrams). Some justification is present, talking about the patterns, with some analysis of how the selections meet the needs of the project. Infrequent contributions (no less than two weeks of sustained contributions) throughout Submission 1 - Submission 2 timeframe.
50pts - An architecture pattern and design patterns are identified, exhibited via non-generic diagrams (diagrams and terms are specific to the application, not generic stock diagrams). Justification, explanations and how the selections align with the established pros or avoid the cons of the patterns, meets the external or internal forces that necessitate such patterns is stated and explained. A true analysis and explanation of the architecture and design for this application/project. Sustained contributions in google doc version history throughout Submission 1 - Submission 2 timeframe.

UX DESIGN

Aesthetics, User research, Personas, UX frameworks, UX architecture and design, UX Mockup

0pts - No UX or minimal UX is presented or evidenced in the project. Minimal or no contributions evidenced, anonymous contributions attributed to student.
10pts - Some UX sections are present and/or minimal description, explanation, and justification of the areas required for UX. Contributions in a small timeframe (work done week before due date as an example).
20pts - All UX sections are present with some explanation of the UX experience for the project. Text/visuals explain, at a high-level, UX work done for the project. Infrequent contributions (no less than two weeks of sustained contributions) throughout Submission 1 - Submission 2 timeframe.
30pts - All UX sections are present and detail a proper explanation and justification of the UX experience for the project. Text/visuals explain the general concepts and UX work done for the project. Infrequent contributions (no less than two weeks of sustained contributions) throughout Submission 1 - Submission 2 timeframe.
40 pts - All UX sections are present and detail a rich and nuanced explanation and justification of the UX experience in totality for the project from install to signup to use of the project. All aspects are explained and shown via text/visuals explaining the concepts and UX work done for this project. Sustained contributions in google doc version history throughout Submission 1 - Submission 2 timeframe.

PROTOTYPE

An animated prototype is presented and exists for the project, submit youtube link of demo and screenshots of it in the google doc, github link if code is used.
0 – No prototype is submitted or displayed

10 – Prototype is a non-animated display, does not accept input to show the flow of the project, infrequent contributions (no less than two weeks of sustained contributions for this grading period)
20 – Prototype accepts input and moves from one screen/menu/visual to the next, showing functionality of the project. Dummy data, sample data are display or used, or real data can be input (not required). Sustained contributions in google doc, figma, canva throughout submission 1 to submission 2.

LATE POLICIES:

Due to the nature of this course, all work will have penalties for late submissions.

- 0 - 48 hours late: 20% Penalty
- 3-4 days late: 40% Penalty
- 5-6 days late: 50% Penalty
- 7+days late: 100% Penalty

ACADEMIC DISHONESTY:

In regard to Academic Dishonesty please review the following link:
http://www.fullerton.edu/senate/publications_policies_resolutions/ups/UPS%20300/UPS%20300.021.pdf

DISABILITIES ACCOMMODATION:

If you have a disability that may qualify you for special accommodations in this course, you should contact: (**http://www.fullerton.edu/dss/**)

Office of Disabled Student Services
Contact: https://www.fullerton.edu/dss/about-us/index.html

Tentative Class Outline:

Week 1(Week of Jan 20th) - First class/Course Introduction/Group Formation
Week 2(Week of Jan27th) – Group Meeting #1
Week 3(Week of Feb3rd) – Group Meeting # 2
Week 4(Week of Feb10th) – **Capstone Pitch Due and Presentations**
Week 5&6(Week of Feb17th&24th) – Group Meeting #3
Week 7&8(Week of Mar3rd&10th) – Group Meeting #4
Week 9(Week of Mar17th) – Group Meeting #5
Week 10(Week of Mar24th) – **Capstone Submission 1 Due and Presentations**
Week 12&13(Week of April7th&14th) – Group Meeting #6
Week 14&15(Week of April21st&28th) – Group Meeting #7
Week 16(Week of May5th) – Group Meeting #8
Week 17 – **Capstone Submission 2 Due and Presentations – Thursday 05/14/26 11:00AM - 12:50PM GH246**